

B.E. ELECTRICAL ENGINEERING SECOND YEAR SECOND SEMESTER EXAMINATION, 2025

ELECTRICAL MACHINES – II

Time: Three hours

Full Marks: 100

(50 marks for each part)

(Use separate Answer Script for each part)

PART - I

1. Answer any four from the following.

4x2.5

(a) "It is not absolutely necessary to laminate the rotor of an induction motor." – True or False? Give reasons.

(b) What will be the direction of rotation of the rotor, if the primary winding is placed on the rotor of a poly-phase induction motor? Give reasons for your answer.

(c) What is the relative speed between the magnetic fields produced by the stator winding and rotor winding of a 3-phase induction motor?

(d) Find an expression for distribution factor of a distributed winding.

(e) Describe the phenomenon of cogging and crawling.

(f) What is deep bar rotor induction motor? How it produces high starting torque?

2. Answer any one from (a) and (b):

10

(a) Explain how a poly-phase winding fed from a balanced poly-phase supply produces a rotating magnetic field with constant amplitude.

(b) Derive an expression for power developed in an induction motor. Also derive the condition for maximum power and draw the power-speed characteristics.

3. Answer any one from (a) and (b):

10

(a) Develop the equivalent circuit of an induction motor from the basic principle.

(b) How the equivalent circuit parameters of an induction motor are determined in the laboratory?

4. Answer any one from (a) or (b):

10

(a) Describe different starting methods of a 3-phase induction motor.

(b) What are the advantages and disadvantages of a SRIM over a SQIM?

[Turn over

5. Answer any one from (a) or (b):

- (a) A 50 kW, 400 volt, 3-phase, 4-pole, 50 Hz wound rotor induction motor develops a maximum internal torque of 240% at a slip of 20% when working under rated voltage and frequency conditions and with slip ring short-circuited.

Determine

- (a) full load slip,
- (b) rotor copper loss at full load and
- (c) starting torque in N-m.

Neglect stator resistance and rotational losses.

- (b) A 15 kW, 3-phase, 4 pole, 50 Hz, induction motor has a full load slip of 0.03.

Mechanical and stray load losses at full load are 3.5% of output power.

Compute

- i) Power delivered by stator to rotor
- ii) Electromagnetic torque at full load
- iii) Rotor ohmic loss at full load.

B.E. ELECTRICAL ENGINEERING SECOND YEAR SECOND SEMESTER EXAM 2025**SUBJECT: ELECTRICAL MACHINES-II**

Time: Three Hours

Full Marks: 100 (50 each part)

Use a separate Answer-Script for each part
PART - II

Answer any three questions. Question no. 3 carries 18 marks.

1. Correct and/or justify following 4x4=16
 - (i) Delta-star is preferable than star-star without neutral three phase transformer.
 - (ii) The cross section of the central limb of a shell type transformer is usually designed to have larger cross section to avoid flux saturation.
 - (iii) The triplen order flux, sometimes links the tank of transformer and produces loss in the same.
 - (iv) Vector group formation is unavoidable with 30 degree phase shift in case of Delta-star or Star-delta transformer.
2. (i) Draw the connection diagram and phasor diagram of the following connections. 9+7=16
 - a) Dd6, b) Yz1 and c) Dz6
- (ii) Show how Dy1 and Yd11 transformers can be connected in parallel successfully.
3. (i) Show with a diagram how 3-rd harmonics current is produced in a transformer. 7+11=18

[Turn over

PART - II

(ii)

In Scott-connected transformers, teaser transformer supplies 0.75 leading power factor load of 25 kW at 220 V and main transformer supplies 0.75 power factor lagging load of 45 kW at 220 V, from a three phase input line voltage of 6600V. Determine the input 3-phase line currents. Draw input current phasors computed for above along with a Scott-connection diagram. Neglect magnetizing currents and the leakage impedance drops.

4. i)

Draw and explain a reactor type on-load tap changer scheme.

7+5+4=16

ii)

Write the merits and demerits of register-type on-load type tap-changers.

iii)

Why high voltage winding of a transformer is preferred for tappings of tap-changer?

5.

Write Short notes on:

8 x 2 =16

(i)

Full wave impulse tests on a transformer.

(ii)

Neutral oscillation in 3-phase transformers & Tertiary windings.